

What are the Advantages and Disadvantages of White Box Testing?

Advantages

- ✓ **Thorough internal coverage** — Tests loops, branches, conditions, and all code paths.
- ✓ **Early bug detection** — Issues are found during development, reducing cost and rework.
- ✓ **Finds hidden issues** — Detects dead code, unreachable code, and security vulnerabilities.
- ✓ **Improves code quality** — Encourages clean, optimized, modular code.
- ✓ **Helps performance optimization** — Identifies inefficient logic and slow loops.
- ✓ **Ideal for critical systems** — Banking, aviation, healthcare, etc.

Disadvantages

- ✗ **Requires strong coding skills** — Testers must understand code, algorithms, and architecture.
- ✗ **Time-consuming** — Analyzing logic and writing detailed test cases takes significant time.
- ✗ **Expensive** — Needs skilled developers/SDETs, increasing project cost.
- ✗ **Cannot detect missing requirements** — Only tests what is written, not what should be written.
- ✗ **Complex for large systems** — Covering all paths in huge applications is difficult.
- ✗ **High maintenance** — Test cases must be updated whenever code changes.

What is Statement Coverage and Path Coverage?

Statement Coverage is a white-box testing technique that checks whether every executable statement in the code has been executed at least once during testing. Its goal is to ensure that no line of code remains untested, helping identify issues like unused statements or simple logical errors. Statement coverage is easy to achieve and gives a basic level of confidence that the code runs without skipping any part.

Path Coverage, on the other hand, is a more thorough technique that ensures every possible execution path in the code is tested. A “path” includes different combinations of branches, loops, and conditions. Because it covers all possible routes the program

can take, path coverage provides deeper validation but is more complex and time-consuming, especially for large applications with many conditional flows.